
UML Diagram 2 — Class Boxes (Fields + Methods)**Shape**

```
+-----+
|           Shape           |
+-----+
| - name : String           |
+-----+
| + Shape(name : String)    |
| + getName() : String      |
| + setName(name : String) : void |
| + toString() : String     |
+-----+
```

Point extends Shape

```
+-----+
|           Point           |
+-----+
| - x : double              |
| - y : double              |
| - z : double              |
+-----+
| + Point(x:double,y:double,z:double) |
| + getX() : double         |
| + getY() : double         |
| + getZ() : double         |
| + setX(x:double) : void   |
| + setY(y:double) : void   |
| + setZ(z:double) : void   |
| + toString() : String     |
+-----+
```

Square extends Point

```
+-----+
|           Square          |
+-----+
| - side : double           |
+-----+
| + Square(side:double,x:double,y:double,z:double) |
| + getSide() : double      |
| + setSide(side:double) : void |
| + area() : double         |
| + toString() : String     |
+-----+
```

Cube extends Square

```
+-----+
|           Cube            |
+-----+
| - depth : double          |
+-----+
| + Cube(depth:double,side:double,x:double,y:double,z:double) |
| + getDepth() : double     |
| + setDepth(depth:double) : void |
| + volume() : double       |
+-----+
```

```
| + toString() : String      |
+-----+
```

Circle extends Point

```
+-----+
|           Circle           |
+-----+
| - radius : double          |
+-----+
| + Circle(radius:double,x:double,y:double,z:double) |
| + getRadius() : double     |
| + setRadius(radius:double) : void|
| + area() : double          |
| + toString() : String      |
+-----+
```

Cylinder extends Circle

```
+-----+
|           Cylinder          |
+-----+
| - height : double           |
+-----+
| + Cylinder(height:double,radius:double,x:double,y:double,z:double) |
| + getHeight() : double      |
| + setHeight(height:double) : void|
| + volume() : double         |
| + toString() : String      |
+-----+
```